SYNTHETIC EVALUATION · SNAPSHOT

The software matches its test examples; real-world accuracy is untested

This report checks a small invented network and hand-checked examples. It does not estimate real outages or prove that AI can understand customer plans.

Synthetic fixtures only. Metrics measure implementation behavior against fixed fixture labels, not physical failure behavior or performance on customer documents.

Dataset and integrity

Measure	Result
Services, facilities and resources / relationship records	43 / 80
Human-verified relationships	78
Data checks	Passed=True · errors=0 · warnings=0

Simple text extraction check

This check asks whether the app found the expected relationship in short fictional sentences. A correct suggestion matches the hand-checked answer; a false suggestion does not; a missed link was expected but not found.

Correct	False	Missed	Precision	Recall	F1	Examples
5	0	0	100%	100%	100%	5

Local deterministic entity/trigger heuristic; LLM path not evaluated. The test uses five short, clear, fictional sentences. Precision asks what share of suggestions were correct; recall asks what share of expected links were found; F1 combines those two scores. This is a parser check, not representative model evaluation.

Human-review state workflow

5 of 5 synthetic status-transition cases matched their expected official inclusion state (100%). Synthetic workflow-state checks only; not user study or human relationship accuracy.

Evidence details on verified relationships

0 of 78 verified fixture edges were missing required metadata (0%). Metadata completeness proxy, not semantic truth of dependency claims.

FIXED SCENARIOS · UNCERTAINTY

All three fictional test answers match

The expected service lists were written down separately, then compared with the app output after each run.

Scenario	Expected	Calculated	Exact match?	Precision	Recall	Time (ms)
substation-alpha-outage	17	17	Yes	100%	100%	0.113
water-pumping-backup-loss	20	20	Yes	100%	100%	0.109
regional-communications-outage	9	9	Yes	100%	100%	0.049

The app matched 3 of 3 expected lists exactly. Expected and calculated counts include the explicitly disrupted starting item. Precision is the share of app-listed items that also appear in the hand-checked list; recall is the share of expected items the app listed. F1 combines precision and recall into one score. Time is shown in milliseconds (thousandths of a second) and is only an informational local measurement.

Redundancy, cycle, and simultaneous disruption checks

- ANY group: dependent remains available while one provider is available; it fails only when all declared alternatives are unavailable.
- ALL group: dependent becomes unavailable when any required provider is unavailable.
- Cycle example: the validator warns about a loop; repeated checks stop when no new services are added, and a disconnected group stays unaffected.
- Unknown/proposed paths: review nodes are returned separately; no unverified edge enters official status propagation.

Uncertainty visibility

Scenario	Open review nodes
substation-alpha-outage	records-archive, shelter-west
water-pumping-backup-loss	records-archive, shelter-west
regional-communications-outage	None

LATENCY · EVIDENCE · LIMITATIONS

A useful software test, not a validated resilience model

The current benchmark verifies code behavior and evidence handling within a small, fixed fixture.

Latency measure	Observed
Repeated deterministic runs	103
Median	0.043 ms
P95	0.054 ms
Environment	Local Python 3.12, NetworkX deterministic fixed-point traversal; latency is informational and hardware-specific.

Do evidence references resolve?

35 evidence reference identifiers were checked and all resolve to graph records (100%). Citation identifiers resolve to graph edges; this does not establish narrative factual correctness.

PDF report completeness

- Impact report required sections present: Yes
- Gaps report required sections present: Yes
- Checklist report required sections present: Yes

Evaluation limits

- No damage, flows, capacities, probabilities, thresholds, durations, restoration, or common-cause behavior are simulated.
- Extraction benchmark is a small synthetic set and excludes ambiguous/adversarial language.
- No local LLM was configured for this run.
- All business and infrastructure statements are fictional.

Next evaluation steps

- Create a customer-approved, independently adjudicated relationship set with ambiguous negatives and contradictory statements.
- Measure owner confirmation precision/recall and source-grounded quote quality, not only workflow state transitions.
- Benchmark graph size and service-layer latency on representative but non-sensitive customer-approved data.
- Conduct a structured review with service owners to test whether paths and proposed actions are understandable and useful.